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Then-Gautier et al.

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(54) **SUPPLEMENTAL ENERGY GENERATION
AND STORAGE FOR TRAINS**

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Sep. 3, 2020, now Pat. No. 11,938,841, which is a
(Continued)

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F16D 55/226 (2006.01)
H02K 7/102 (2006.01)

(52) **U.S. Cl.**
CPC **B60L 7/10** (2013.01); **F16D 55/226**
(2013.01); **H02K 7/102** (2013.01); **B60L**
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(58) **Field of Classification Search**
CPC B60L 7/10; B60L 2220/44; F16D 55/226;
H02K 7/102

See application file for complete search history.

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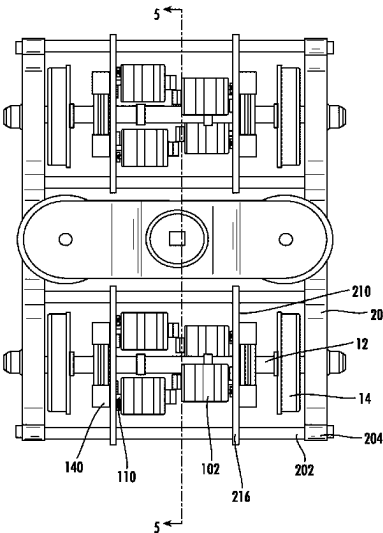
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Samotny Ltd.

(57) **ABSTRACT**

The invention relates to the supplemental generation of
energy from operation of a train, and specifically to the
generation of energy in connection to the rotation of disc
brake rotors in combination with generators. Rotation of the
disc brake rotors creates rotational energy that is transmitted
to the generators, which then transmits the energy to a series
of batteries for storage. The batteries may be stored in the
platform for the train and/or within the train car itself.
Energy from the batteries may be utilized by removal of the
batteries from the train or through a number of outlets,
sockets or connectors associated with the train car or plat-
form.

19 Claims, 17 Drawing Sheets



Related U.S. Application Data

continuation-in-part of application No. 16/876,010, filed on May 16, 2020, now abandoned, and a continuation-in-part of application No. 16/871,821, filed on May 11, 2020, now abandoned, said application No. 16/876,010 is a continuation of application No. 16/726,862, filed on Dec. 25, 2019, now abandoned, said application No. 16/871,821 is a continuation of application No. 15/858,848, filed on Dec. 29, 2017, now Pat. No. 1,066,814.

- (60) Provisional application No. 63/024,888, filed on May 14, 2020, provisional application No. 62/784,904, filed on Dec. 26, 2018, provisional application No. 62/440,775, filed on Dec. 30, 2016.

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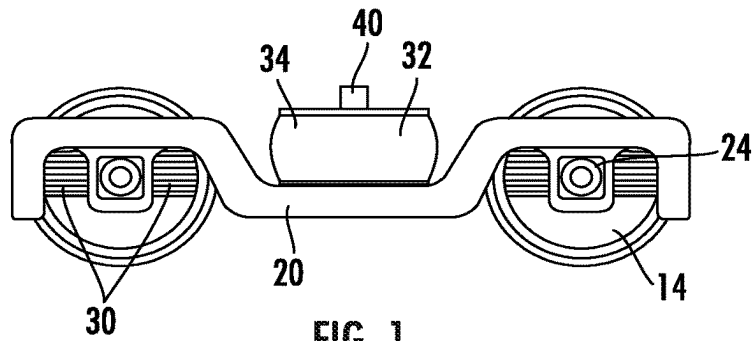


FIG. 1
PRIOR ART

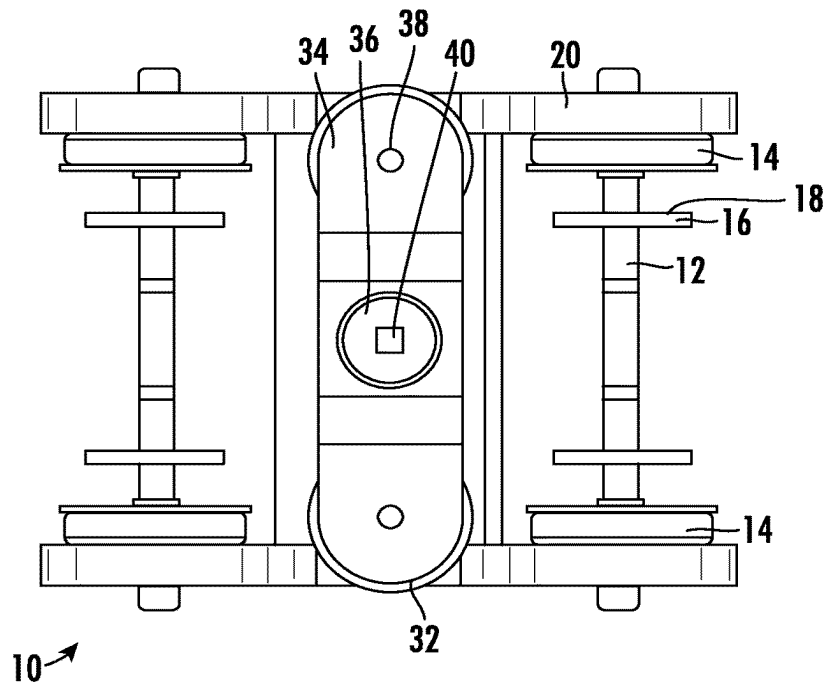


FIG. 2
PRIOR ART

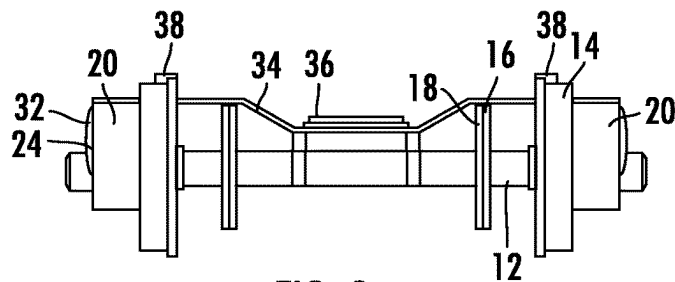


FIG. 3
PRIOR ART

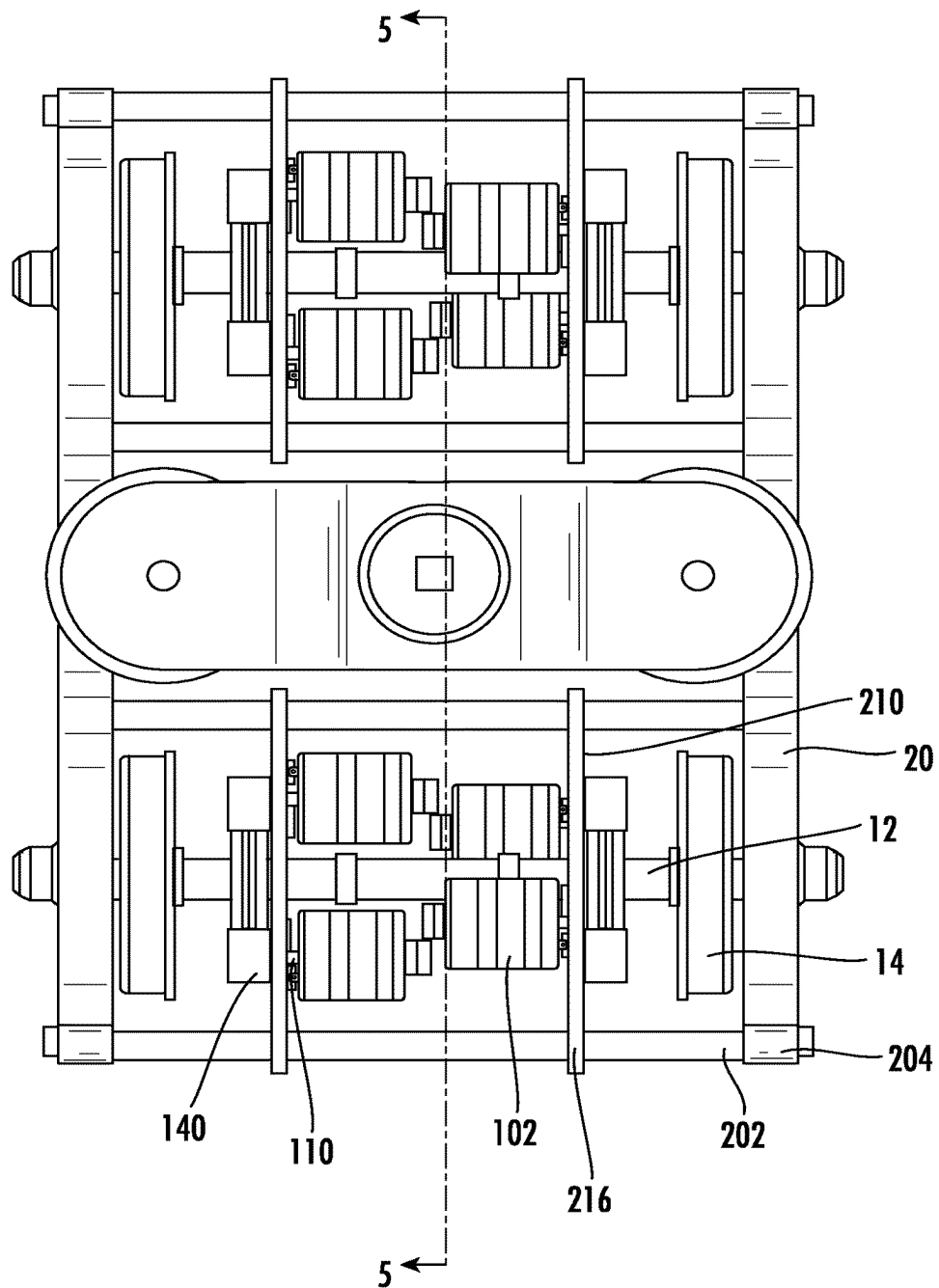


FIG. 4

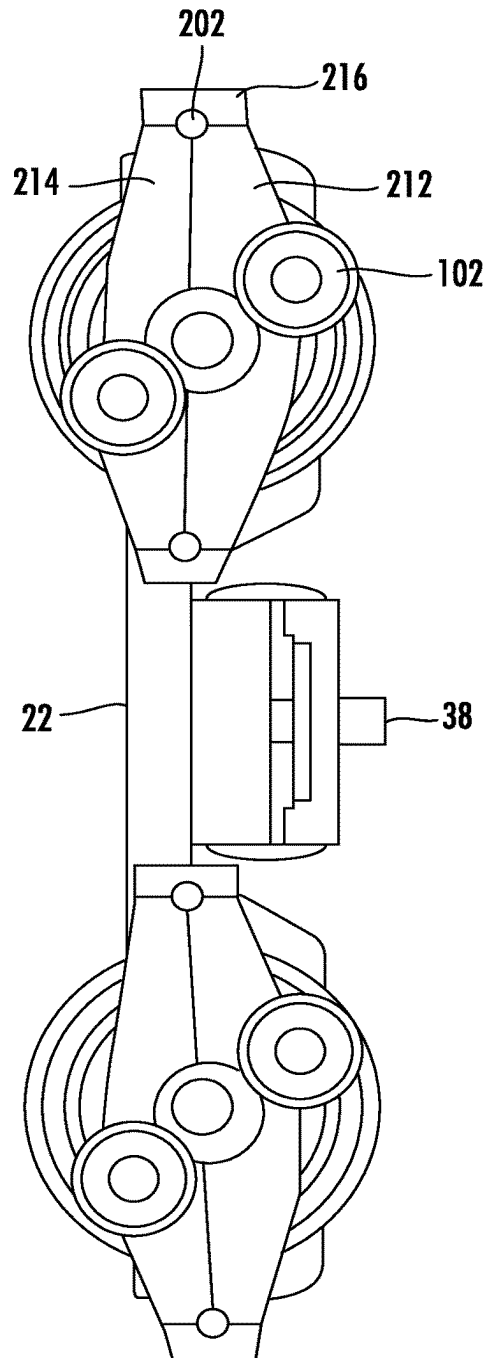


FIG. 5

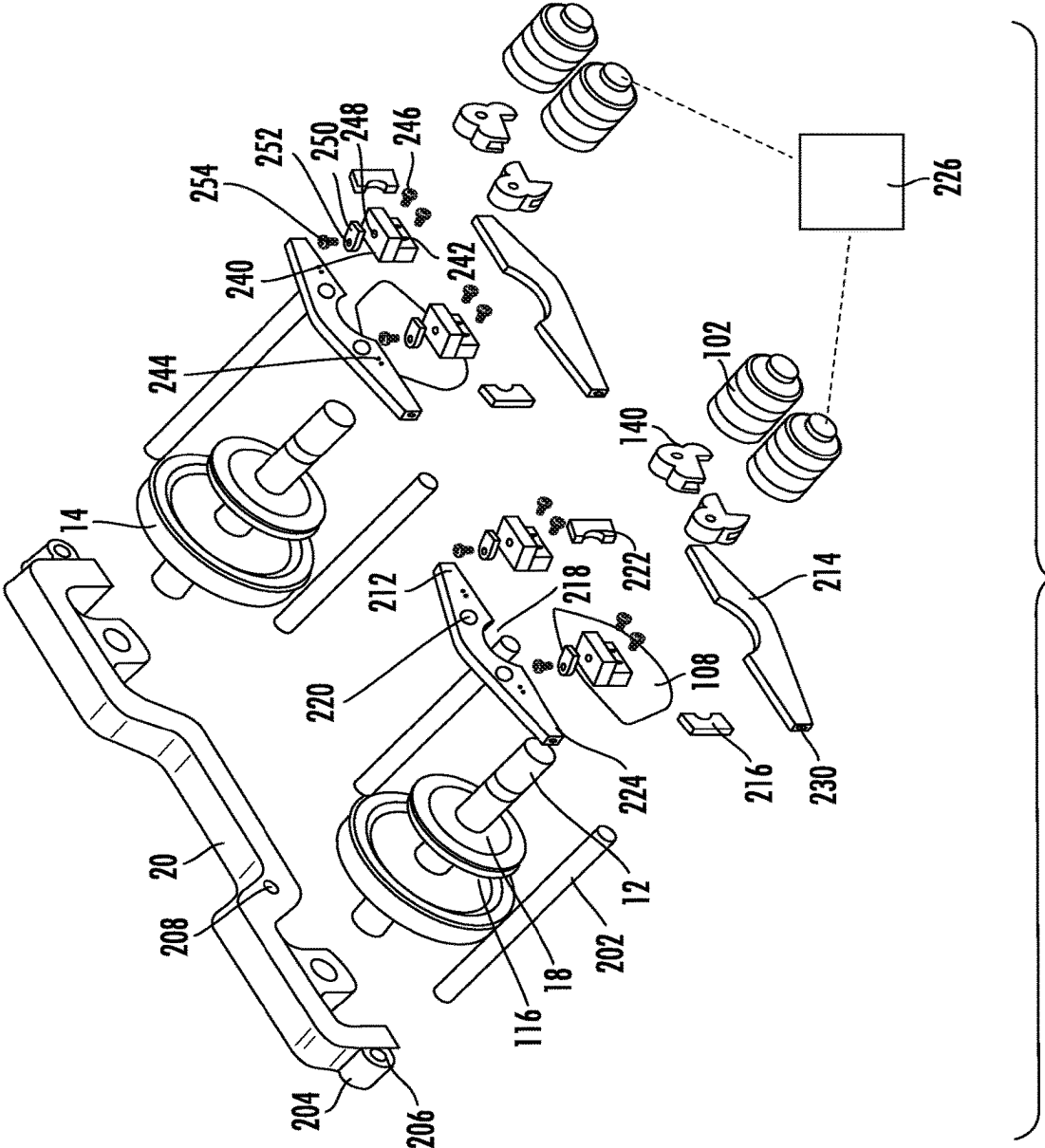
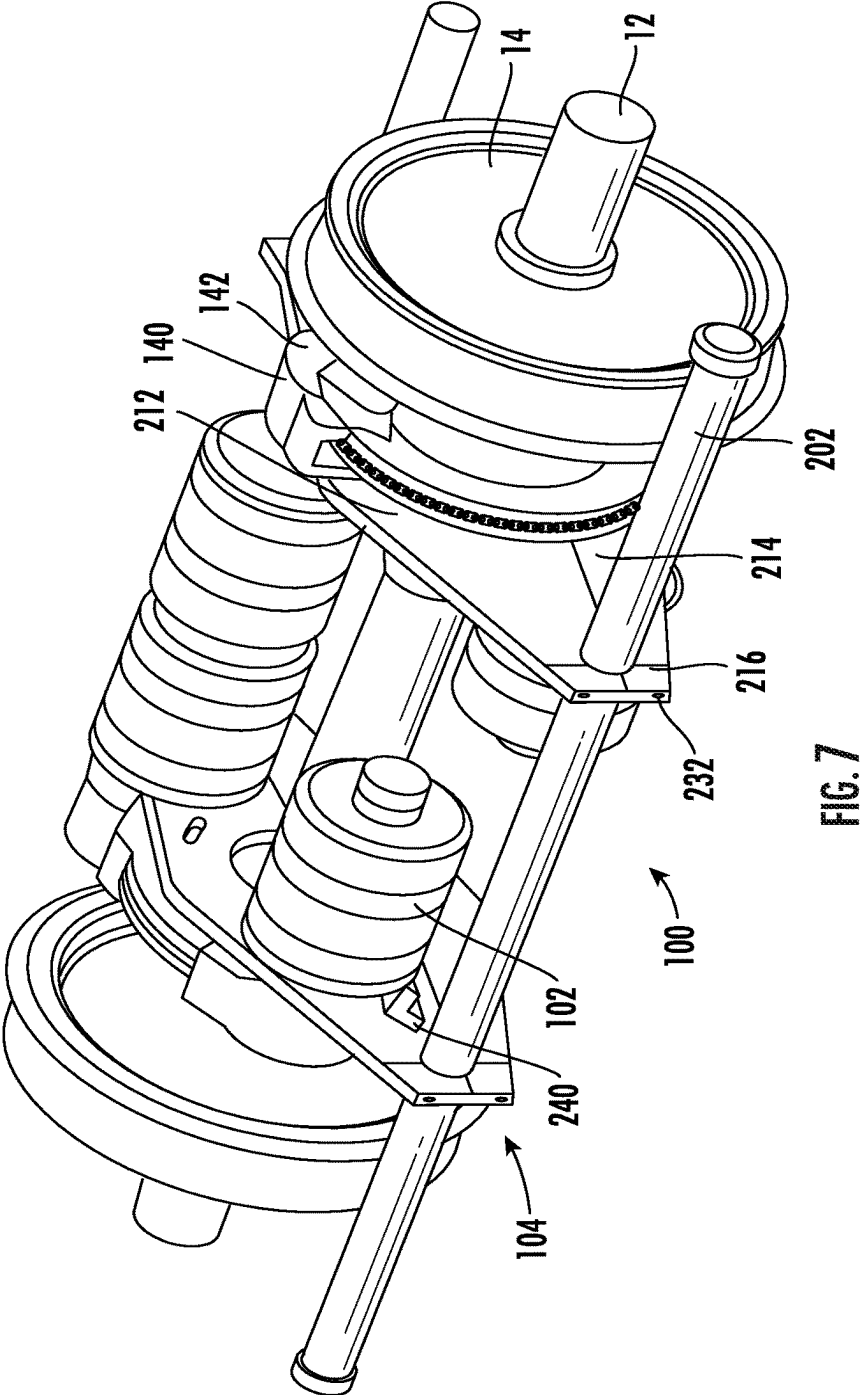


FIG. 6



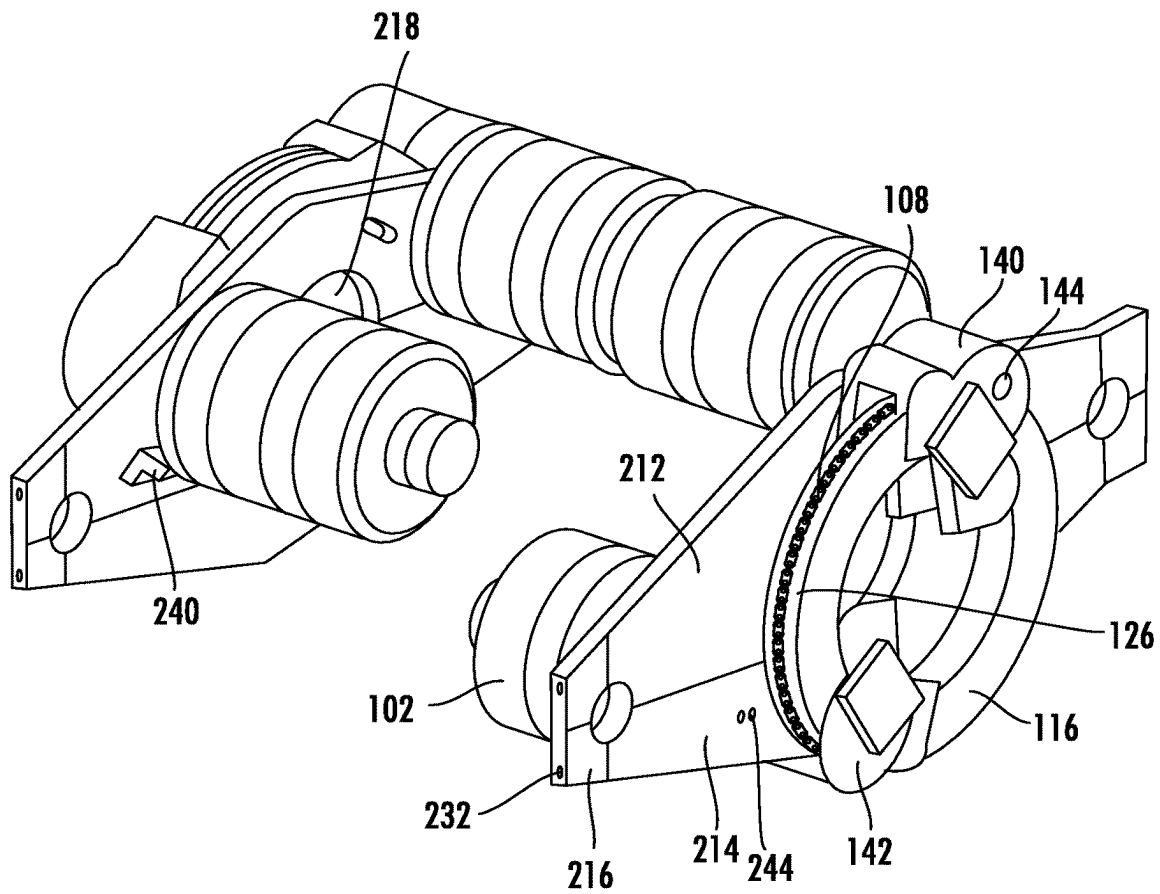


FIG. 8

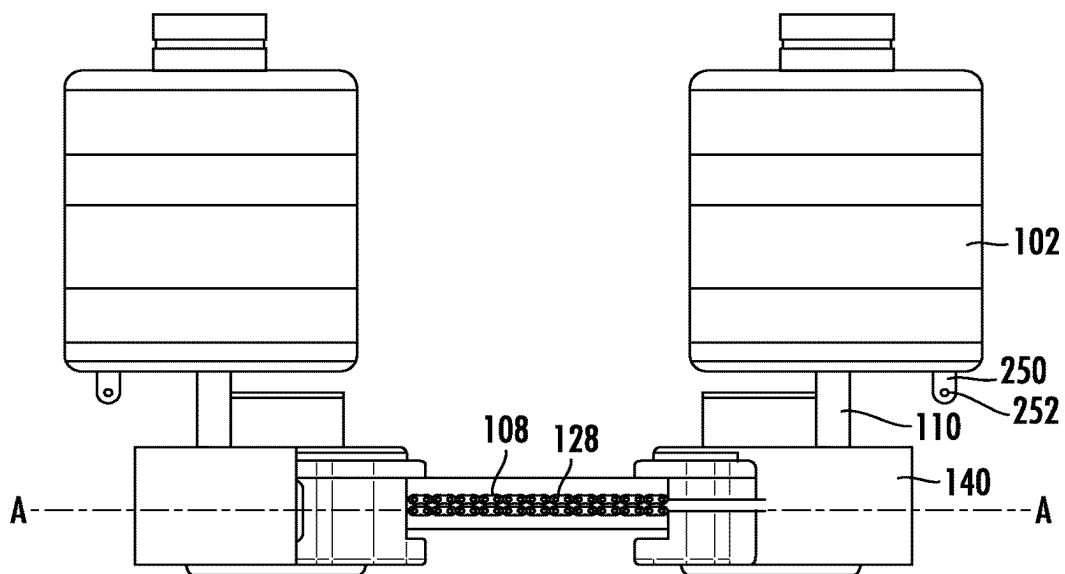


FIG. 9

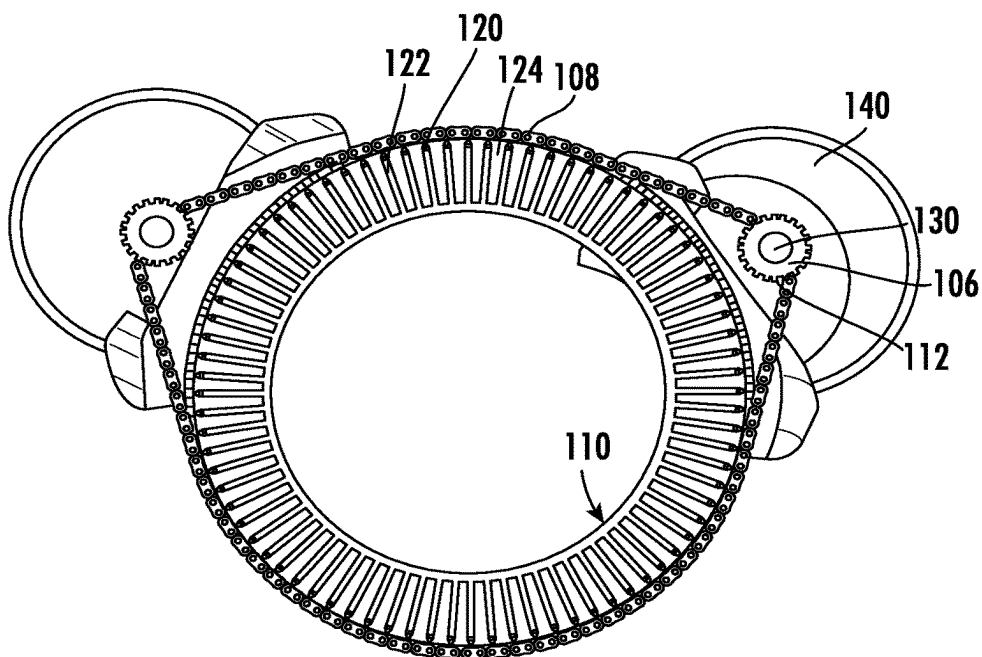


FIG. 10

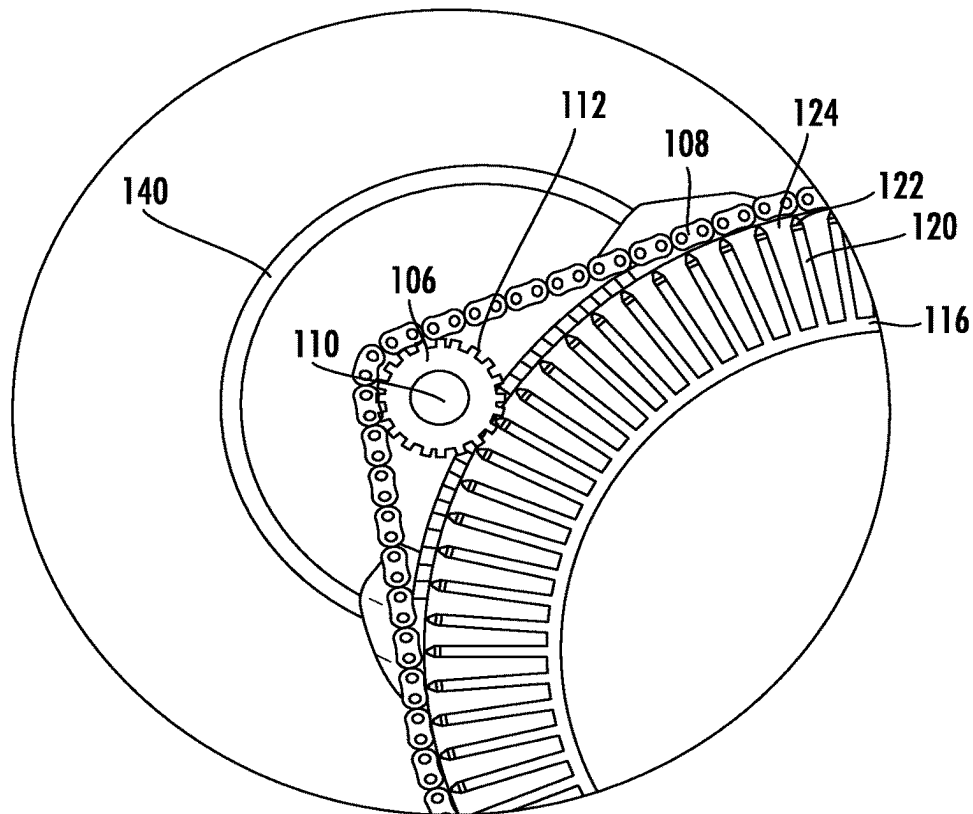


FIG. 11

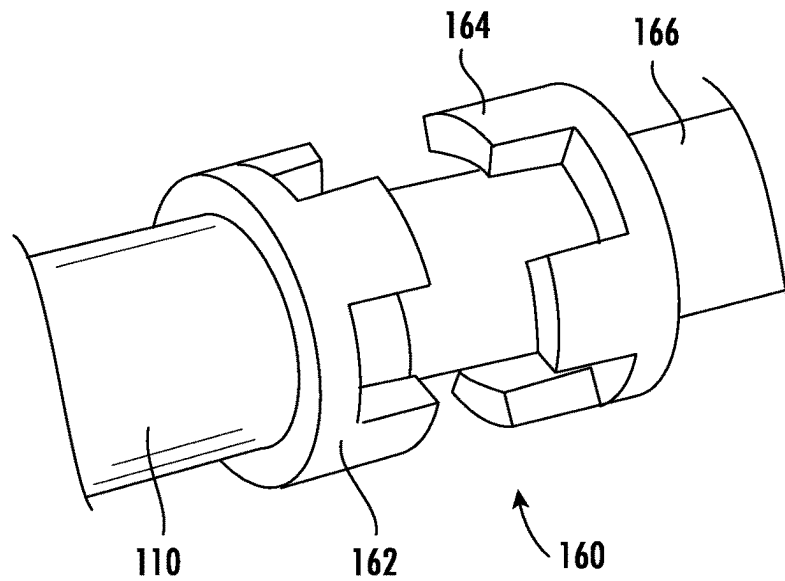


FIG. 12

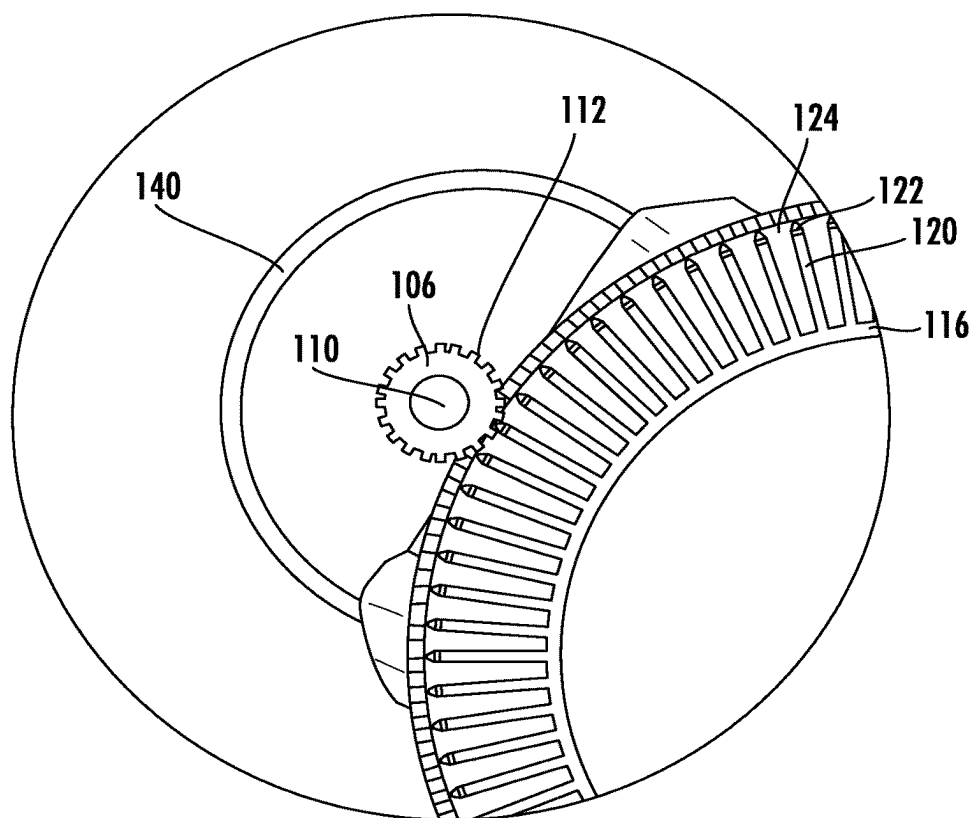
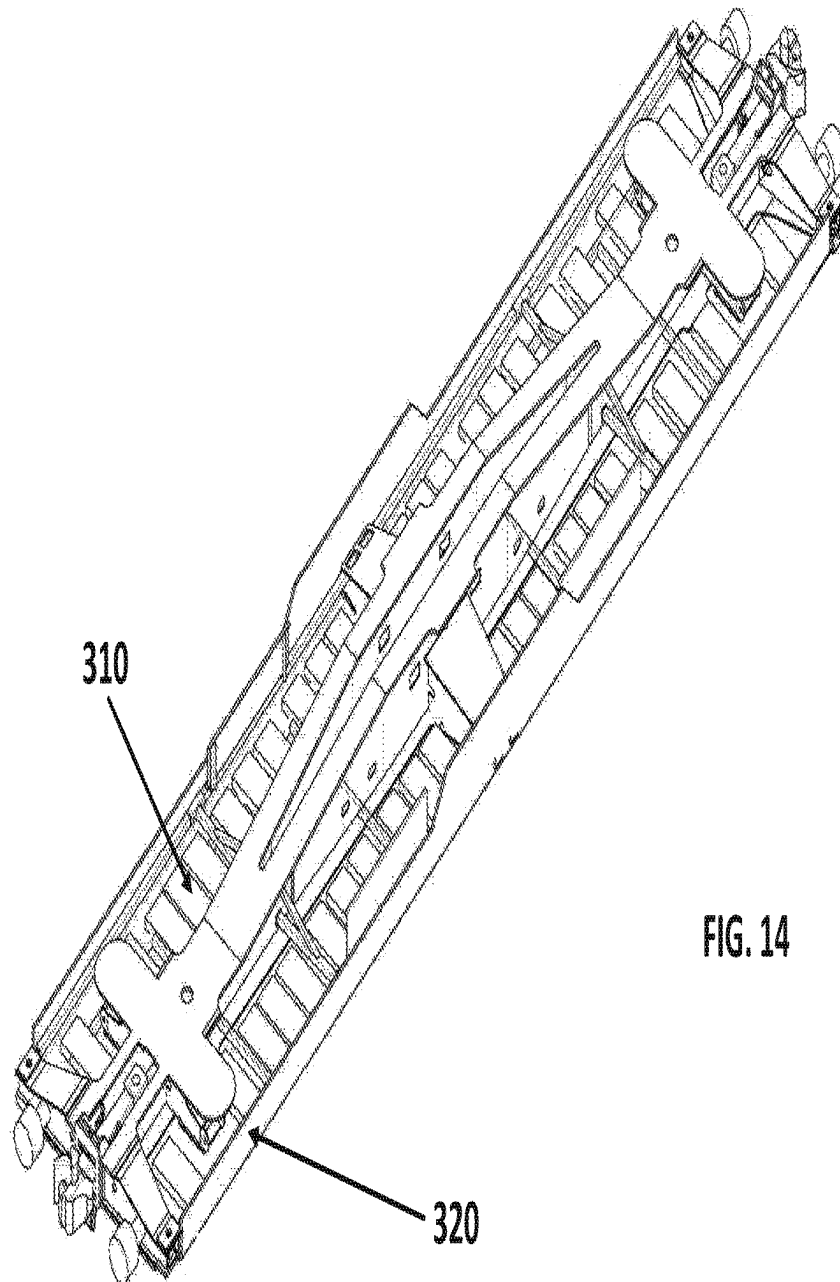
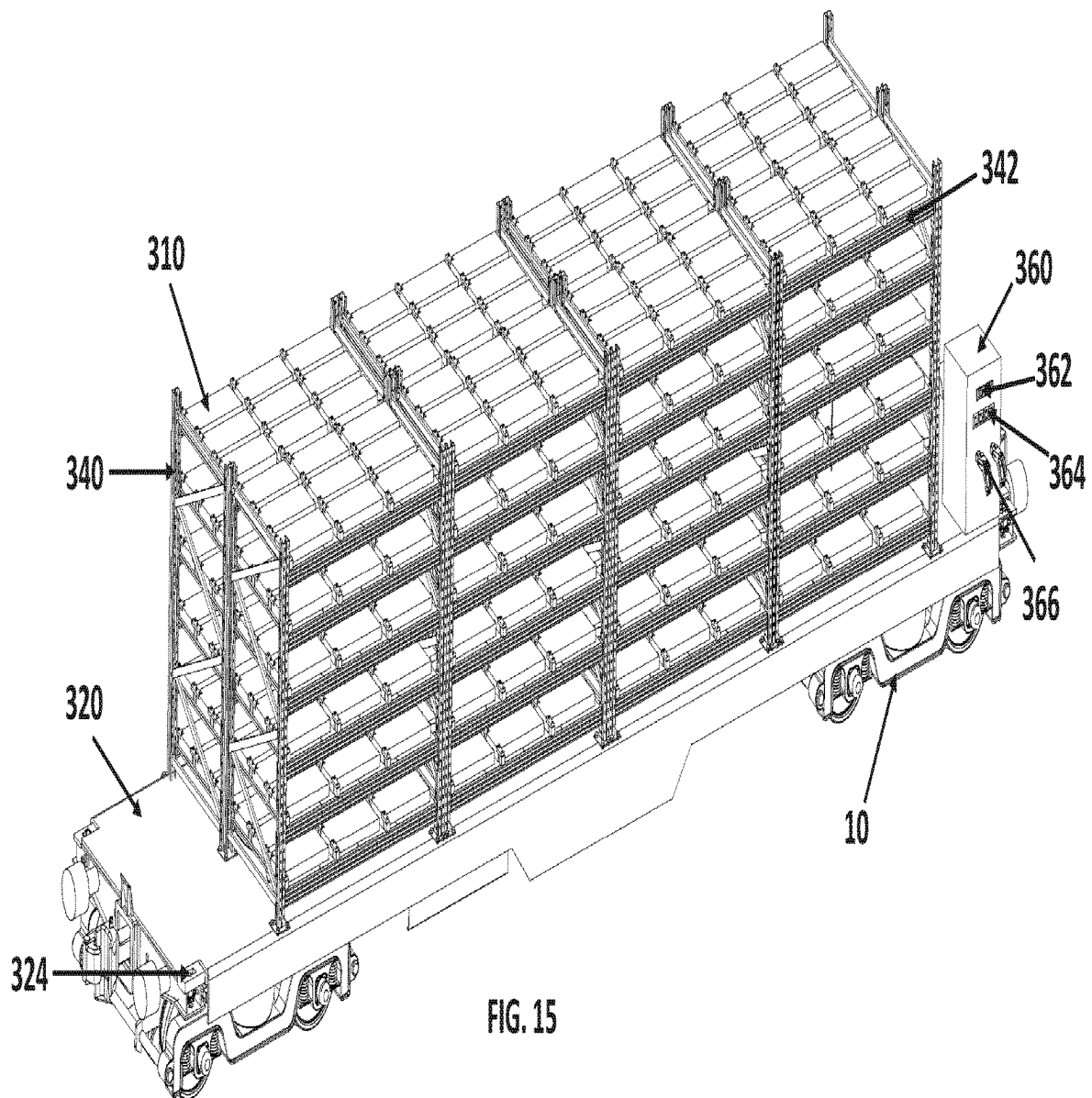
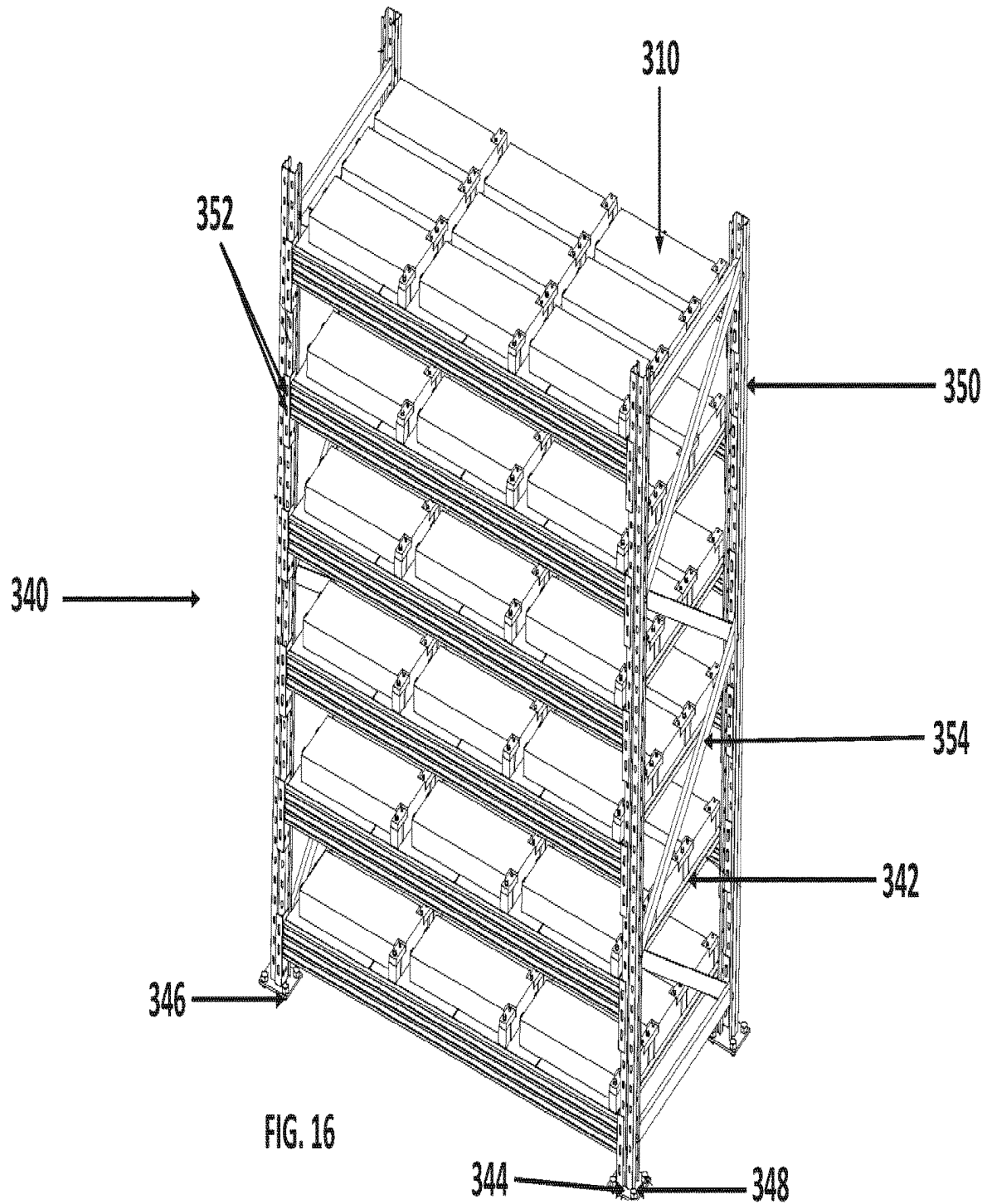


FIG. 13







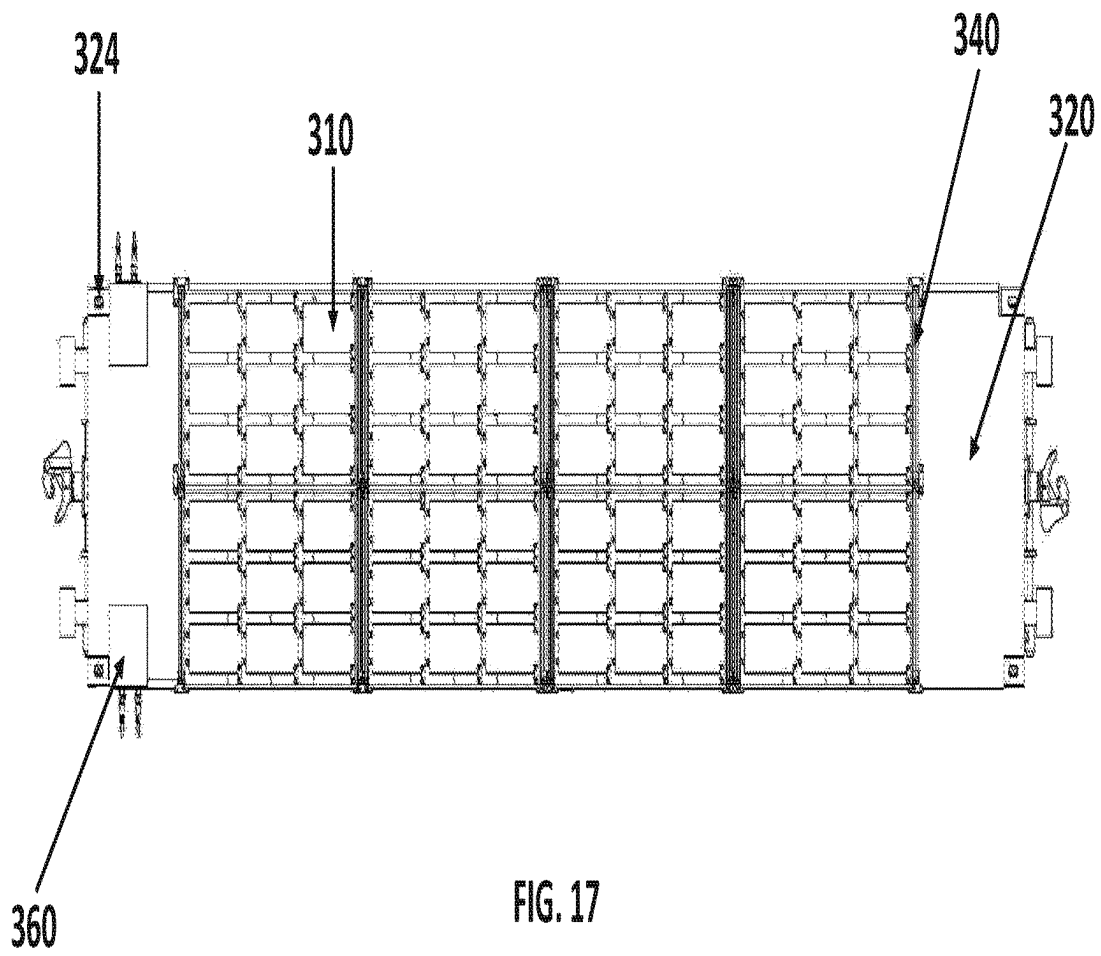
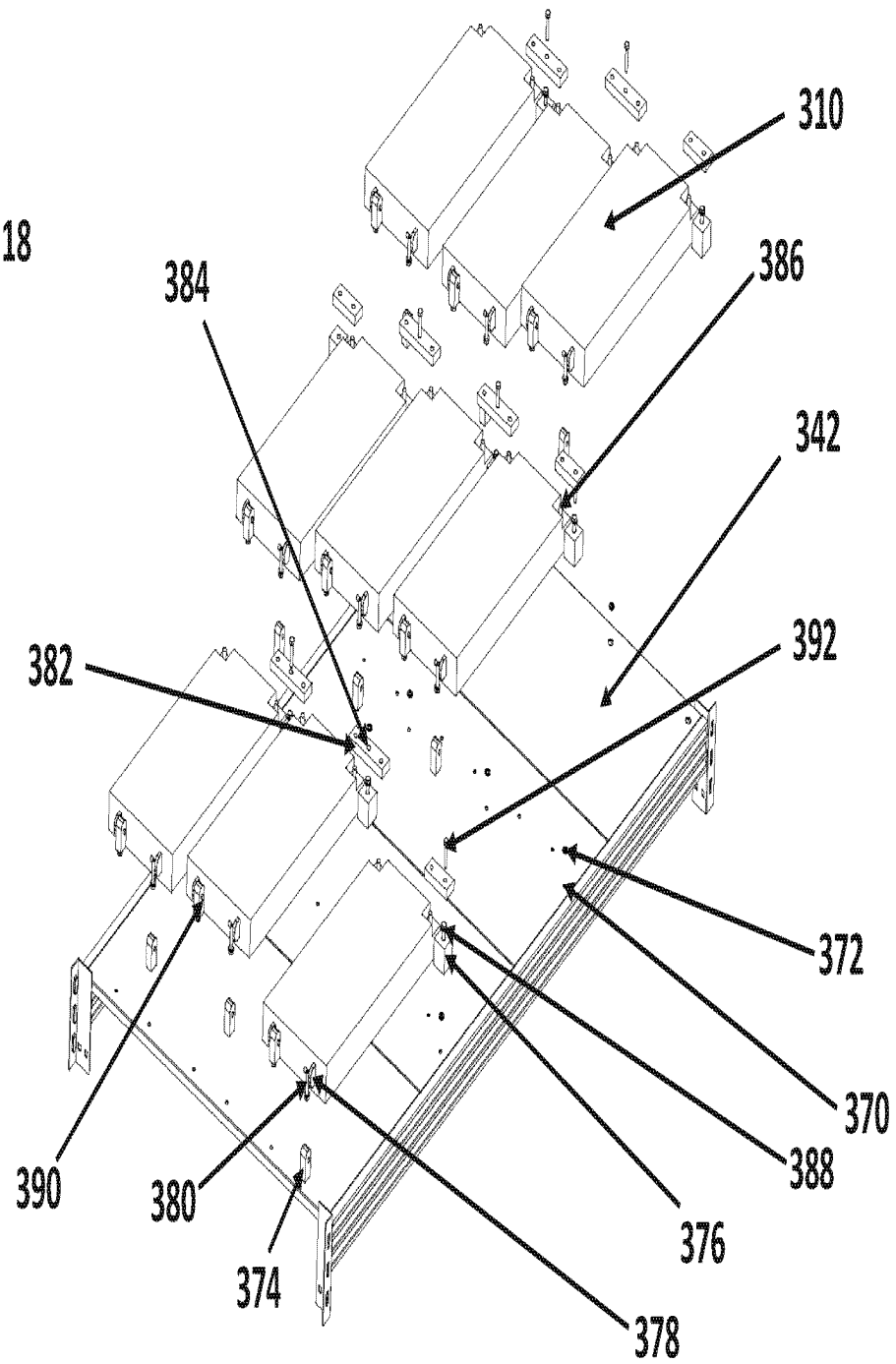
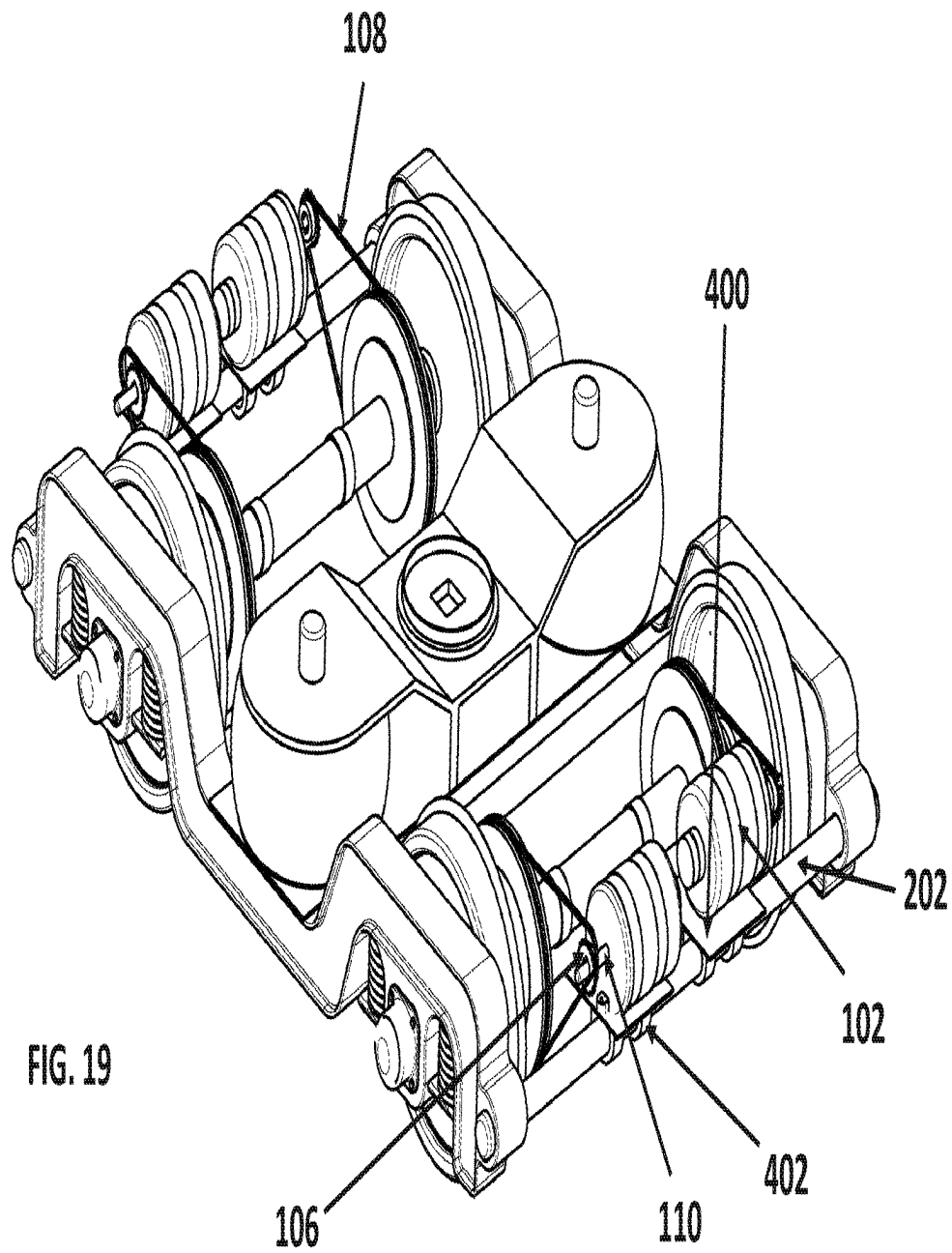


FIG. 18





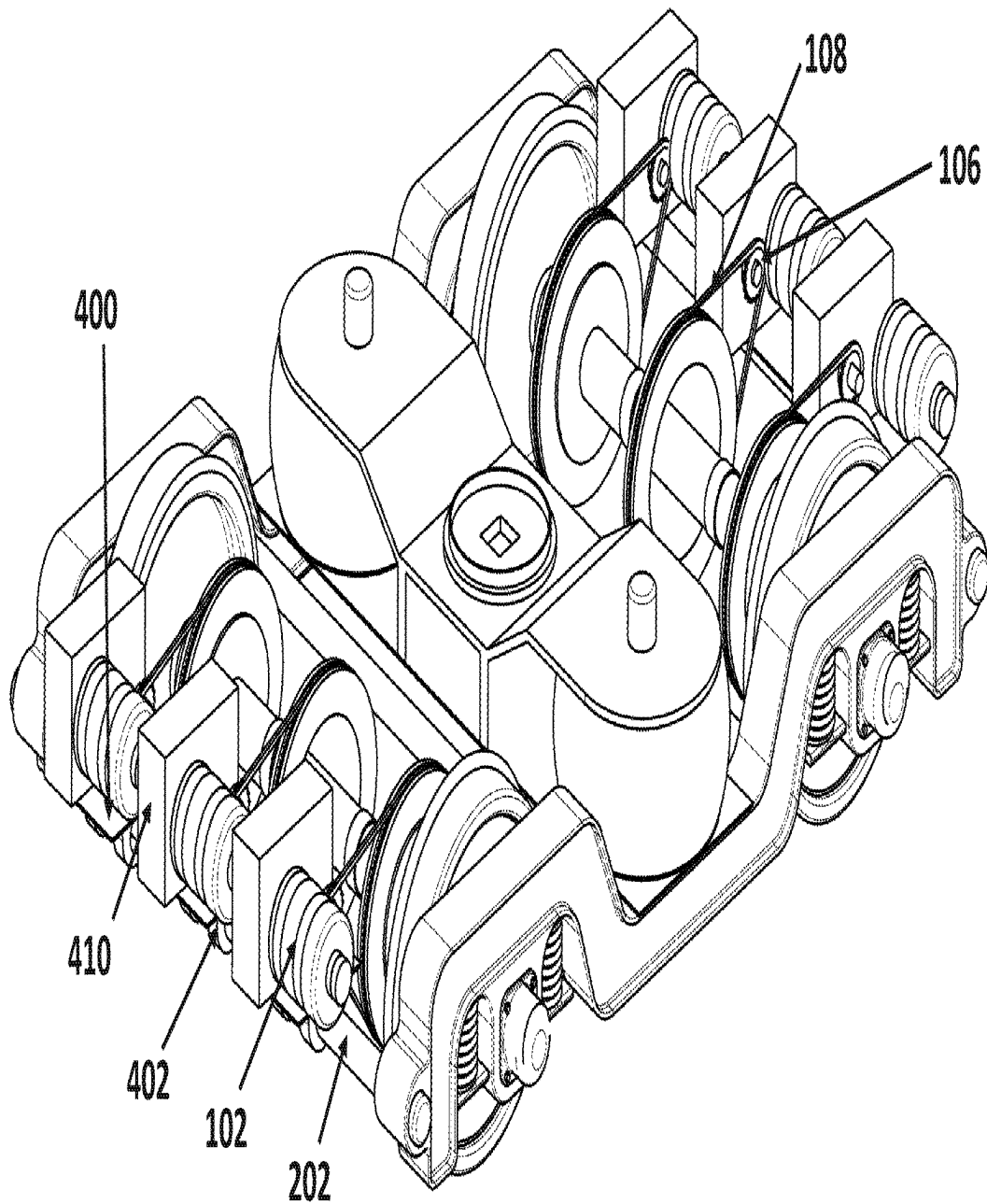
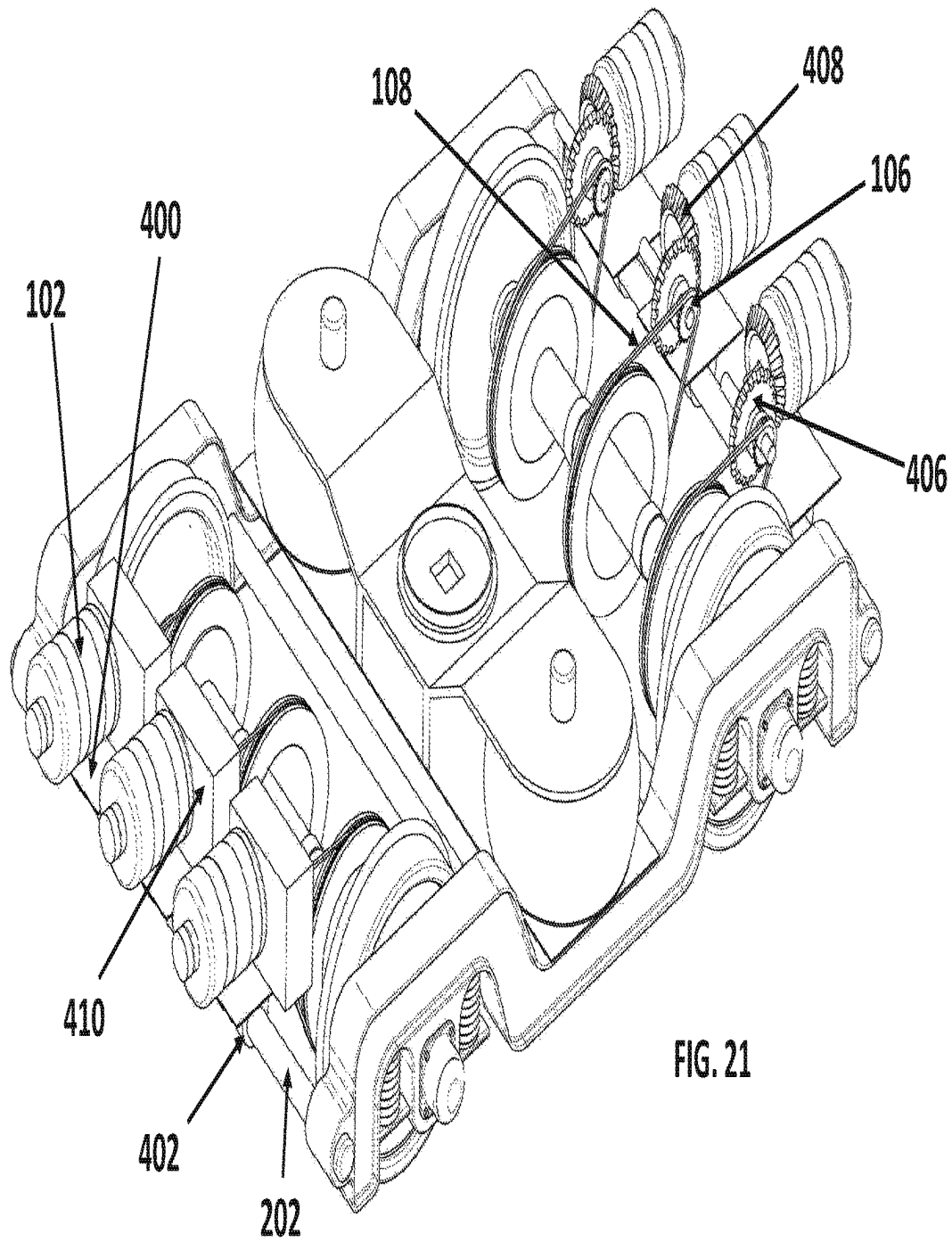


FIG. 20



SUPPLEMENTAL ENERGY GENERATION AND STORAGE FOR TRAINS

This application is a continuation of U.S. non-provisional patent application Ser. No. 17/011,891, filed Sep. 3, 2020, and which issued on Mar. 26, 2024 as U.S. Pat. No. 11,938,841, which is a continuation-in-part of U.S. non-provisional patent application Ser. No. 16/871,281, filed on May 11, 2020, and which issued on Nov. 1, 2022 as U.S. Pat. No. 11,485,235, which is a continuation of U.S. non-provisional patent application Ser. No. 15/858,848, filed on Dec. 29, 2017, and which issued on Jun. 2, 2020 as U.S. Pat. No. 10,668,814, which claims benefit to U.S. provisional patent application Ser. No. 62/440,775, filed on Dec. 30, 2016, the disclosures of which are herein incorporated by reference in their entirety. This application is also a continuation of U.S. non-provisional patent application Ser. No. 17/011,891, filed Sep. 3, 2020, and which issued on Mar. 26, 2024 as U.S. Pat. No. 11,938,841, which claims priority to U.S. provisional patent application No. 63/024,888 filed on May 14, 2020, the disclosure of which is herein incorporated by reference in its entirety. This application is also a continuation of U.S. non-provisional patent application Ser. No. 17/011,891, filed Sep. 3, 2020, and which issued on Mar. 26, 2024 as U.S. Pat. No. 11,938,841, which is a continuation-in-part of abandoned U.S. non-provisional patent application Ser. No. 16/876,010, filed on May 16, 2020, which is a continuation of abandoned U.S. non-provisional patent application Ser. No. 16/726,862 filed on Dec. 25, 2019, which claims benefit to U.S. provisional patent application Ser. No. 62/784,904 filed on Dec. 26, 2018, the disclosures of which are herein incorporated by reference in their entirety.

The following documents and references are incorporated by reference in their entirety, Erlston et al (U.S. Pat. Appl. No. 2008/0078631) and Bodenstein et al (U.S. Pat. Appl. No. 2012/0091724).

FIELD OF THE INVENTION

The present invention relates to the supplemental generation of energy from operation of a vehicle, and specifically to the generation and storage of energy from the movement of a train in combination with generators for converting rotational energy into electrical energy for storage and subsequent use.

BACKGROUND OF THE INVENTION

The rail system in the United States alone comprises about 140,000 miles of rail for use to transport freight and passengers. The vast network of railways provides an efficient way to move freight (both imports and exports) around the continental United States. Statistics from 2017 indicate that trains deliver 5,000,000 tons of freight and 85,000 passengers per year. As of 2018, railways accounted for about 28% of all freight movement by ton-miles and more than 500,000 miles of passenger travel.

Trains are also generally a more energy efficient mode of freight transportation than trucking and other means with statistics showing that freight trains can average about 470 miles per freight-ton. As a result, despite the higher percentage, moving freight by rails only accounts for about 2% of transport-related emissions. For these reasons, among others, federal forecasts indicate that freight traffic is likely to increase substantially over the next several decades.

Despite all this, railway companies still face difficulties and obstacles in being successful. Among other things, as the rails are privately owned by the rail companies, the costs to maintain them can be substantial. According to recent studies, freight railroads spend more than forty cents on the revenue dollar on capital expenditures and maintenance (each accounting for about one half of the cost or about twenty cents on the dollar). This is significantly higher than other industries and manufacturers, where capital expenditures average about 3% of revenue.

As for passenger rails such as Amtrak and Metra, most passenger rail companies do not own most of the stations and rail they utilize. There are also significant costs in maintenance of equipment. As a result, there are often very low, if any, profit margins and many passenger carrying train systems and companies are highly subsidized and struggle to economically compete with other forms of personal transportation. As an example, Amtrak has lost money every year since its formation in 1971.

While trains are a more environmentally friendly means of transportation, in view of global environmental concerns regarding other types of energy, there is still a need and demand to do more to utilize cleaner energy and reduce emissions.

Accordingly, it is an object of the present invention to provide an effective and efficient system and method for generating and storing power seamlessly as a byproduct from the operation of trains.

Another object of the present invention is to provide a way to utilize native operation of trains to generate and store power to generate income.

It is another object of the present invention to provide a way to utilize trains to generate and store environmentally friendly power.

These and other objects and features of the present invention will become apparent in view of the present specification, drawing and claims.

SUMMARY OF THE INVENTION

The present invention relates to the generation of energy through operation of a train. In one embodiment, the invention comprises an energy generation system that is operationally attached to the axles in the train bogie. The energy generation system utilizes a motion transmission mechanism (chain, V-belt or notched timing style belt, continuously variable transmission, mechanical gearing or a gearbox or a fluid based torque converter) that engage disc brake rotors or any other rotating members attached to the train's axles to transmit rotational energy to generators through the use of sprockets and associated shafts. The generator then transmits that energy to an energy storage unit such as a plurality of batteries. In an alternate embodiment, the sprockets directly engage the disc brake rotors. A bracket may be utilized to retain the sprocket in place relative to the disc brake. Alternatively, the sprocket may be positioned over and/or attached to the disc brake caliper.

The batteries may be stored beneath or within the train wagon or platform and/or in the train cars themselves. Racks may be utilized to allow batteries to be stacked vertically to increase the amount of storage therein. Energy stored in the batteries may be utilized by removing the batteries after they are charged or through adapters or connectors on the train cars to allow equipment or vehicles to be connected thereto for charging.

In order to help retain the generators in place within the train bogie, a support frame may be utilized. The support

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frame may comprise a pair of shafts or rods that extend between opposing bogie side frame members. A traverse frame member is attached or connected to the pair of shafts and extends parallel to the disc brake rotor. The traverse frame member may comprise a top frame member, a lower frame member and a pair of end members that are attached together to allow the axles to extend through a central hole and freely rotate therein. The frame members include opening that are sized to rotationally receive the shaft. Bushings may be utilized to facilitate rotation of the shafts in the openings. Bracket members on or attached to the generators and frame members allow the generators to be removably connected to the frame.

Additional aspects and advantages of the invention are further described below, parts of which will become apparent from the descriptions below or be learned from the practice of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and/or additional aspects and advantages of the invention shall be obvious and easily understood from the following descriptions of the embodiments in combination with the appended drawings, where:

FIG. 1 is a side elevation view of a basic train bogie with a pair of axles.

FIG. 2 is a top plan view of the train bogie and wheels of FIG. 1.

FIG. 3 is a front elevation view of the train bogie and wheels of FIG. 1.

FIG. 4 is a top plan view of the energy generation systems of the present invention installed on the train bogie of FIG. 2.

FIG. 5 is a cross-sectional view of the energy generation system and train bogie taken along line 5-5 of FIG. 4.

FIG. 6 is a partial exploded view of the energy generation system and train bogie of FIG. 4.

FIG. 7 is a front perspective view of the energy generation system of the present invention with a pair of wheels on an axle.

FIG. 8 is a perspective view showing components of the energy generation system of the present invention shown in FIG. 7.

FIG. 9 is a top plan view of components of the energy generation system of the present invention.

FIG. 10 is a cross-sectional view of the components of the energy generation system of FIG. 9.

FIG. 11 is a cross sectional view showing another embodiment of components of the energy generation system of the present invention showing the sprocket directly engaging the disc brake rotor.

FIG. 12 is a partial perspective view of one embodiment of a clutch for use with the energy generation system of the present invention.

FIG. 13 is a cross sectional view showing another embodiment of components of the energy generation system of the present invention showing the sprocket directly engaging the disc brake rotor and the use of a chain.

FIG. 14 is bottom perspective view of one embodiment of a storage arrangement for batteries with a train platform.

FIG. 15 is a perspective view of another embodiment of a storage arrangement for batteries utilizing racks.

FIG. 16 is a perspective view of one embodiment of a rack for the storage of batteries.

FIG. 17 is a top plan view the embodiment shown in FIG. 16.

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FIG. 18 is a partial exploded view showing a plurality of batteries attachable to one shelf of a rack.

FIG. 19 is a perspective view showing another embodiment of the energy generation system of the present.

FIG. 20 is a perspective view showing another embodiment of the energy generation system of the present.

FIG. 21 is a perspective view showing another embodiment of the energy generation system of the present.

The preceding description and drawings merely explain the invention and the invention is not limited thereto, as those of ordinary skill in the art who have the present disclosure before them will be able to make changes and variations thereto without departing from the scope of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

While the present invention is susceptible of embodiment in many different forms, there is shown in the drawings and will herein be described in detail specific embodiments, with the understanding that the present disclosure is intended as an exemplification of the principles of the present invention and is not intended to limit the invention to the embodiment disclosed.

Referring to FIGS. 1 through 3, basic components from a train bogie 10 as presently embodied in the prior art are shown in connection with a pair of axles 12 each with a pair of wheels 14 for engaging the rail (not shown). On each axle 12 are a pair of discs 18 that allow disc brake rotors 16 to be attached thereto for stopping or slowing the train when calipers are engaged. The bogie 10 itself comprises a pair of side frame members 20 that are connected together by a bogie transom 22. The axles 12 are rotationally connected relative to the bogie side frame members 20 through axle boxes 24.

A pair of springs 30 proximate each wheel extend between the bottom of the axle box 24 and the underside of the side frame member 20 to act as the primary suspension for the train bogie 10 and what it is carrying. Secondary suspension members in the form of air springs 32 are shown on the bogie transom 22. A bolster 34 extends over the air springs 32 and includes a center plate 36 with a central pivot member 40 to connect to a railcar or platform. Pins 38 may extend from atop of the bolster 34 to facilitate connection with the railcar 322 or platform 320. While one embodiment of a train bogie is shown and disclosed, it is appreciated that other train bogies having varying parts or different numbers of axles and wheels may be used and not depart from the scope of the invention.

Referring now to FIGS. 4 through 10, one embodiment of the energy generation system 100 of the present invention is shown as having two pair of disc brake rotors 116, four pair of generators 102, support frames 104, mechanical couplers 106, chains 108 and connecting members 110. The energy generation system 100 is designed to fit within the frame of the train bogie for use with existing or new bogies. While the energy generation system is shown and disclosed as used in connection with a disc brake for each wheel, it is appreciated that it may be utilized with selected disc brakes on the bogies and not depart from the scope of the present invention. The energy generated may go to an energy storage system 310 for subsequent use.

Referring to FIG. 10, one embodiment of the disc brake rotor 116 shows it as being annular in shape with slots 124 circumferentially extending therethrough. The disc brake

rotor **116** is sized and shaped to fit over and be attached to the disc **18** on the axle **12** so that it will rotate with the rotation of the wheels **14**.

The mechanical coupler **106** is shown as a sprocket or gear having a plurality of teeth **112** spaced apart its circumference and sized to engage corresponding openings **128** in a roller chain **108** that is placed around the edge of the disc brake rotor **116**, facilitating the rotation. In order to facilitate the mechanical connection of the chain **108** to the disc brake rotor **116**, the spokes or elongated members **120** forming the slots **124** have tapered ends **122** for engaging the openings **128** in the links of the chain **108**. Furthermore, while a chain is shown and disclosed, it is appreciated that other rotation conveyors or motion transmission mechanisms such as, but not limited to, V-belts or notched timing style belts, continuously variable transmissions, mechanical gearing or a gearbox or a fluid based torque converter may be utilized. While tapered ends are shown and disclosed, it is appreciated that the chain or other rotational conveyor may engage the disc brake rotor **116** through friction or other known connectors or connecting methods. Among other things, various connectors may be fixed or positioned within a channel **126** extending about the circumference of the disc brake rotor **116**. Further, while a chain **108** is shown and disclosed, as shown in FIG. **13**, it is appreciated that the mechanical coupler **106** may directly engage the slots **124** formed in the disc brake rotor **116** without the need for a chain. It is also appreciated, as shown in FIG. **11**, that a chain **108** may be utilized to facilitate rotation of the mechanical coupler **106** with the disc brake rotor **116**.

While a disc brake is shown and disclosed, it is appreciated that the energy generation system of the present invention may be used with other rotating members that could be attached or coupled to the train's axle (i.e., rotate with the wheels) including, but not limited to, other circular or annular members attached to the axles. If other components are used, it is appreciated that they could be customized to engage a mechanical coupler or permit a chain or other rotational conveyor to wrap around and engage the component to provide for the generation of energy as disclosed herein.

The mechanical coupler **106** is connected to the connecting member, shown as a shaft **110**, through its center. In order to maintain the coupler **106** in position relative to the disc brake rotor **116**, a bracket **140** may be used. Similar to the disc caliper in operation, the bracket **140** remains parallel to the disc brake rotor **116** surface. In order to not interfere with the operation of the disc brake, the coupler **106** and brackets **140** are placed apart from the caliper about the disc brake rotor **116**.

Referring again to FIG. **7**, the bracket **140**, having a hollow interior defined by a pair of side walls **142**, is positioned astride of the disc brake rotor **116**. Opposing holes **144** extend through the side walls **142** for receiving a portion of the shaft **110** therethrough to allow the coupler **106** to be aligned with the slots **124**. It is appreciated that the bracket may only have a hole **144** on one side wall. Bushings may be inserted into the holes **144** to facilitate rotation of the shaft **110**. While a separate bracket is shown and disclosed, it is appreciated that to conserve space or provide for ease of installation, the caliper may act as one of the brackets for the sprocket, either for the sprocket itself or to permit the chain to pass through to engage the sprocket.

The shaft **110** is mechanically connected to the generator **102** for generating energy in the form of either alternating current ("AC") or direct current ("DC"). In one embodiment, the generators are alternators. The energy is then

transmitted and/or stored in an energy storage component or system **310** on or in the train car, platform or bogie, or in the vehicle's energy system. This may include feeding into the alternator/battery bus, and/or into a separate vehicle electronics circuitry. While two generators are shown per disc brake, it is appreciated that one or more generators may be utilized per disc brake.

Referring now to FIG. **12**, the energy generation system **100** may be equipped with a clutch **160** to only generate energy when desired or needed. The positive clutch **160** shown in the figures includes a jaw member **162** on the end of the shaft **110** and a corresponding jaw member **164** positioned on a driven shaft **166** for the generator **102**. When the clutch **160** is engaged, the corresponding jaw members **162,164** are engaged to allow for the transmission of the rotational movement to the generator **102**. When disengaged, the shaft **110** will rotate without transmitting energy to the generator **102**. Similarly, the generation of energy may be electronically shunted, so the load is minimal when not generating energy.

In order to maintain the generators **102** in place relative to the disc brake rotor **116**, a support frame **104** may be utilized. Referring to FIGS. **4**, **6** and **7**, one embodiment of the support frame **104** is shown having a pair of rods **202** extending between the side frame members **20** of the train bogie frame. The rods **202** may be connected to the side frame members **20** through corresponding brackets **204** having holes **206** therein, and/or through holes **208** on the interior portion of the side frame members **20** themselves. While brackets and holes are shown and disclosed, it is appreciated that the rods or other connecting members may be attached to or relative to the side frame member through other known means.

A pair of traverse frame members **210** extend proximate and parallel to the respective disc brake rotors **116** and are attached or otherwise connected to the rods **202**. In order to facilitate installation, the traverse frame members **210** include an upper frame member **212**, a lower frame member **214** and a pair of end members **216**. Each of the upper and lower frame members **212,214** has a corresponding concave central opening **218** to permit the axle **12** to extend therethrough and freely rotate. Bushing members may be utilized in the central concave opening to facilitate rotation and protect the upper and lower frame members.

Circular openings **220** extend through one or more of the upper and lower frame members **212,214** to accommodate portions of the shafts **110** and allow them to freely rotate therein. Bushings may be utilized to facilitate the rotation of the shafts and protect the frame members from abrasion and wear. The openings are offset to not interfere with the brake disc caliper or other train components. It is appreciated that the openings may be moved within or between the frame members depending upon the particular structure of the train bogie and the train components therein.

Each of the end members **216** has an opening **222** sized and shaped to receive the rod **202** or a portion therethrough. If the rods **202** are circular or arcuately shaped, or if desired for rods of other cross-sectional shape, a corresponding opening formed by cut-away portions **224** in the opposing corners of the upper and lower frame members **212,214** may be utilized. Once the frame members **212,214,216** are positioned about the rod **202**, they may be attached to one another through bolts, rivets or other connectors through corresponding holes **230** and **232** therethrough. It is appreciated that the frame members may also be attached to each other and/or the rods or connected using welding or other known means. While shown as separate pieces, it is appre-

ciated that the upper and lower frame members may have openings therethrough for receiving and attaching to the rods, wherein they may be positioned prior to installation of the rods in the train bogie.

The generators **102** may be attached to the support frame **104** to provide support for the energy generation system **100**. One embodiment showing the support frame attaching to the generators is shown in FIG. **6** as having L-shaped brackets **240** having a pair of holes **242** for attaching to frame members **212,214** through corresponding holes **244** using bolts **246**. While the bracket is shown as L-shaped and attached using bolts, it is appreciated that it may be alternatively sized and shaped and may be integral with the end members or otherwise connected thereto using other known means.

A bracket or connecting member **250** may be integral with or otherwise attached using bolts, welding or other known means to the generator **102**. The bracket **250** includes a hole **252** that aligns with hole **248** to allow a bolt to pass therethrough to attach the brackets **240, 250** and thus the generator **102** and frame members **212,214** together.

It is further appreciated that depending on the structure of the train bogie, the generator may be offset from the location of the disc brake rotor. One example of the generator **102** being offset from the disc brake rotor **116** is shown as having a plate or base member **400** for holding or attaching to the generator **102**. The plate **400** may be attached to rod **202** through brackets **402**.

Referring to FIG. **20**, the system may also include a transfer case **410** to allow the generator **102** to be mounted at an angle from the sprocket **106** and shaft **110** if desired. FIG. **21** illustrates the transfer case **410** and the generator **102** attached thereto being perpendicular to the brake disc rotor **116**.

The transfer case **410** also may include gearing to modify the drive ratio as desired. A continuous variable transmission or fixed gear transmission also may be utilized to alter the drive ratio as the change changed speed. An example of a gear system for use with the energy generation system is shown as having a first bevel gear **406** that rotates about shaft **410** and engages bevel gear **408** that rotates about shaft **110** to generator **102**. It is appreciated that different sizes, types and numbers of gears may be used depending on the space and desired output.

In operation, one or more tapered ends **122** of the spokes **120** in the disc brake rotor **116** engages openings in the chain **108**. As the disc brake rotor **116** is rotated, the chain **108**, and thus the sprockets **106** and shafts **110** will also rotate as the teeth **112** of the sprockets **106** are moved out of and into engagement with the openings **128** in the chain **208**. Thus, as the wheel **14** moves, the disc brake rotor **116** moves, and the shafts **110** going to the generators **102** move thereby creating energy to be stored in the train. If desired, the clutches **160** may be utilized to selectively permit control of the transmission of the energy to the respective generators **102**.

It is appreciated that the ratio between the size of the sprockets **106** and the disc brake rotor **116** or other rotational component may vary to increase or optimize energy generation. As an example, one ratio between the diameter of the disc brake rotor **116** for which the chain **108** is wrapped around to the sprocket or coupler **106** is 1:9, wherein when the disc brake rotor **116** rotates once, the sprocket **106** and shaft **110** attached thereto will rotate nine times. This ratio may be employed to increase energy generation for trains that travel at a higher speed (e.g., between 35 mph and 65 mph). It is appreciated other ratios will have different ideal

ratios, defined by the variables including the type of generator used and the average speed of the train.

Energy generated by the energy generation system **200** may be sent to one or more energy control units or electrical converters that then transmit the energy to a plurality of batteries **310** located within compartments or storage areas on the train car **322** and/or wagon or platform **320**. The electrical converter and system regulates the electrical output based on vehicle condition, battery charge levels, battery or device draw on the battery and ambient conditions to enhance energy generation. While a variety of different batteries may be utilized, one example of a battery that may be used with the present invention is a Panasonic® battery with a 5.3 kWh storage capacity.

FIGS. **14** through **17** illustrates various embodiments of battery storage arrangements for trains. As shown in the figures, it is appreciated that the batteries **310** may be stored within or on the underside of a train car platform **320** and/or within the train car **322** itself. FIG. **14** illustrates one embodiment of a plurality of batteries **310** attached or positioned on the underside of the train platform or wagon **320**. While twenty-one batteries are shown on each side for a total of forty-two batteries, it is appreciated that a different number and arrangement of batteries may be utilized depending on the particular wagon and batteries utilized and not depart from the scope of the present invention.

The wagon **320** includes a plurality of twist locks **324** about its corners to allow it to be releasably attached to train bogies **10** having the energy generation systems **100** of the present invention. The platform may then be loaded with cargo or be part of a train car for carrying passengers or batteries. Examples of different types of containers that may be placed on top of the platform include, but are not limited to, hoppers, gondolas, flatcars, boxcars, reefers and tanker cars.

FIGS. **15** through **18** illustrate one embodiment of batteries being stored in racks **340** within the train cars **322**. The racks **340** may include any number of levels or shelves **342** for storing batteries **310** in a variety of configuration including, but not limited to, horizontally along the width and or depth of the shelves.

The racks **340** may include feet **344** having a plurality of holes **346** therein to allow them to be bolted or otherwise secured to the top of the wagon **320**. In order to increase storage capacity, the shelves of the rack may be positioned at different levels to provide tiers having various heights. Vertical posts **350** having a plurality of spaced openings **352** allow the shelves **342** to be adjusted to different heights as needed or desired to accommodate the batteries **310**. Cross members **354** may extend along the sides and back of the rack between the posts **350** to provide support to the racks **340**.

The example of a rack shown in FIG. **16** includes a plurality of shelves closely placed together to allow batteries to be stored flat as shown in FIG. **16**. It is appreciated that the arrangement and height of the shelves may vary depending on a number of factors including the internal height of the train car and the particular dimensions and arrangement of the batteries.

The racks may be arranged in rows to permit multiple racks to be stored within a train car to increase or maximum battery storage capacity. Referring again to the embodiment shown in FIG. **15**, a platform and/or train car could accommodate two rows of four racks each having six shelves of nine batteries for a total of four hundred and thirty-two batteries. If the storage capacity of a battery is 5 kWh that would amount to a total of 2,160 kWh of storage. It is

appreciated that a single train car may store batteries within the car itself and/or the platform.

In order to retain the batteries 310 in place during movement of the train, they may be secured or attached to the shelves 342 of the rack 340 when stored. The shelving may include a plurality of spaced apart openings 370,372 that allow a series of blocks or retaining members 374,376 to be attached thereto through screws, bolts or other connectors. Corresponding retaining members or walls 378 extend from one end of the batteries 310 and include a pin or other connector 380 for removably inserting in corresponding opening in retaining member 374. The other end of the battery may include cut away portions at its corners that allows a plate or connecting member 382 having a plurality of holes 384 spaced apart thereon to attach to pin or post 386 extending thereto and a bolt or other connector 392 for connecting to retaining member 376. As shown, the plate 382 may include two or three holes 384 depending on whether it is attaching a single corner of a battery 310 or opposing corners of adjacent batteries 310.

It is appreciated that the train cars may include an energy distribution system that connects to the batteries to allow the stored power to be transmitted to other batteries, equipment or machines. The energy distribution system may include connectors and adapters for transferring power from the batteries 322 to the other batteries, equipment or machines. Referring again to FIG. 15, the adapters and connectors 362,364,366 may be included in panels 360 that may fit within the side of the train cars 322. The adapters and connectors may include but are not limited to 120V sockets 362, 220V sockets 364 and quick charging adapters 366 for fast charging of electric vehicle batteries. It is also appreciated that the batteries may be removed and replaced after charging for remote use.

A software application may be utilized to track the amount of energy being produced and stored in the batteries. The amount of energy that is consumed from the train from the generated energy may also be tracked to monitor and charge for its usage if desired. A telemetric device may be used to remotely transmit the information collected about the stored and used energy.

The preceding description and drawings merely explain the invention and the invention is not limited thereto, as those of ordinary skill in the art who have the present disclosure before them will be able to make changes and variations thereto without departing from the scope of the present invention.

The invention claimed is:

1. A system for the generation of energy in a train having disc brakes on axles for wheels within the train bogie, the system comprising:

a disc brake rotor having a periphery; a generator; a support frame for positioning the generator within the train bogie; a coupler member positioned adjacent to the disc brake rotor; and a rotational conveyor for mechanically connecting the generator to the disc brake, the rotational conveyor extending around at least a portion of the periphery of the disc brake rotor and the coupler member, and wherein energy may be continuously generated upon rotation of the wheels.

2. The system of claim 1 wherein the generator comprises a connecting member that mechanically connects the generator to the coupler member.

3. The system of claim 2 wherein the connecting member is a shaft.

4. The system of claim 2 which further comprises a bracket that maintains the coupler member in position relative to the connecting member of the disc brake rotor.

5. The system of claim 1 wherein the rotational conveyor comprises a chain.

6. The system of claim 1 wherein the rotational conveyor comprises a belt.

7. The system of claim 1 which further comprises a bracket that maintains the coupler in position relative to the connectors of the disc brake rotor.

8. The system of claim 1 which further comprises a clutch, the clutch selectively controlling the generation of energy by the generator.

9. The system of claim 1 where the support frame comprises a pair of rods extending parallel to the axle and a traverse frame member extending between the two rods proximate to the disc brake rotor.

10. The system of claim 9 wherein the traverse frame member comprises an upper frame member, a lower frame member and a pair of end members.

11. The system of claim 10 wherein the generator comprises a shaft that mechanically connects the generator to the coupler member and wherein at least one of upper frame member and the lower frame member comprises an opening to rotatably receive the shaft.

12. The system of claim 1 which further comprises an energy storage system.

13. A system for the generation of energy in a train having disc brakes on axles within the train bogie, the system comprising:

a disc brake rotor having a periphery; a generator; a support frame for positioning the generator within the train bogie, the support frame comprising: a pair of rods extending parallel to the axles; and a traverse frame member extending between the two rods proximate to the disc brake rotor; a coupler member positioned adjacent to the disc brake rotor; a rotational conveyor disposed about at least a portion of the periphery of the disc brake rotor and the coupler member; and a connecting member mechanically connecting the generator to the coupler member.

14. The system of claim 13 which further comprises a bracket that maintains the coupler in position relative to the disc brake rotor.

15. The system of claim 13 which further comprises a clutch, the clutch selectively controlling the generation of energy by the generator.

16. The system of claim 13 wherein the traverse frame member comprises an upper frame member, a lower frame member and a pair of end members, wherein at least one of the upper frame member and the lower frame member comprises an opening to rotatably receive the connecting member.

17. The system of claim 13 which further comprises a clutch, the clutch selectively controlling the generation of energy by the generator.

18. The system of claim 13 wherein energy may be continuously generated upon rotation of the axles.

19. The system of claim 18 which further comprises a clutch, the clutch selectively controlling the generation of energy by the generator.